

Step-by-step guide:

to support

**From Framework to Practice: Operational tools for Identifying
Subpopulations and Evolutionary Significant Units to Support
Conservation Management**

(DOI: to be filled in)



Subpopulation and ESU delineation framework by the
IUCN Conservation Genetics Specialist Group (CGSG)

How to cite: to be filled in

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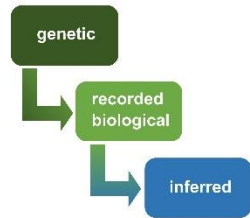
Introduction

The purpose of this guide is to help users navigate a two-phase webtool to apply a framework for identifying subpopulations and Evolutionarily Significant Units (ESUs) in species based on different data types (genetic, recorded biological and inferred; Figure 1A). It first explains the general features of the webtool and then presents two case studies to demonstrate how the tool can be applied.

Overview of the testing process/framework (Figure 1B)

1. Species are assessed in so-called ‘candidate scenarios’ (see below for more details)
2. Within the framework three different lines of evidence (“genetic evidence”, “recorded-biological evidence” and “inferred evidence”) are used to accommodate different data availabilities for species (Figure 1A)
3. Four different scenarios can be assessed simultaneously, although the first scenario is always the null hypothesis (either a single Metapopulation or a single ESU) formally tested against other candidate scenarios as a baseline.
4. In Phase 1, genetically meaningful subpopulations are identified (Figure 1B)
5. Candidate scenarios scoring ≥ 2 points qualify for Phase 2
6. In Phase 2, the subpopulations identified in Phase 1 may be tested individually or grouped into candidate ESU scenarios (Figure 1B). As with Phase 1, different lines of evidence are used to assess their ESU status.
7. Please note: Genetic or genomic data are required in Phase 2 to achieve a score of ≥ 10 and delineate genetically meaningful ESUs.

a) Lines of evidence



b) Two - Phase Framework

Phase 1: Identify subpopulations



Score candidate **subpopulation scenarios**

***Null - Hypothesis:** Single genetic **metapopulation** (no intraspecific subdivision)
→ one ESU (i.e. no testing of Phase 2 necessary)

Candidate Scenarios & Scoring:

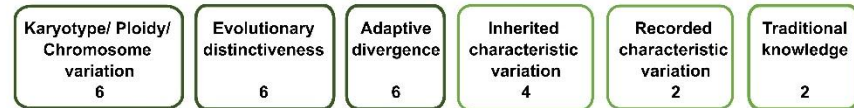
Subpop - Scenario A) x subpopulations
Subpop - Scenario B) n subpopulations
...

Scoring: **≥ 2 points** is required to
proceed to **Phase 2**
(group into 'meaningful' ESU - Scenarios)

Candidate Scenarios & Scoring:

ESU - Scenario X) k ESUs
ESU - Scenario Y) j ESUs
...

Phase 2: Delineate Evolutionarily Significant Units (ESUs)



Score candidate **ESU scenarios** (note: these scenarios can differ from the subpopulation scenarios)

***Null - Hypothesis:** 1 ESU

Scoring: **≥ 10 points** is required to delineate **distinct ESU(s)**
< 10 points due to various situations:

- a) final score could not reach or exceed 10 points (even if all missing data were supportive of the tested scenario)
= no support for the ESU - Scenario
- b) final score does not currently reach or exceed 10 points*, but could do so if missing data subsequently prove to be supportive of the scenario
= scenario indicates possible ESUs (pESUs)

*if only non-genetic evidence is currently scored, the ESU-scenario should be rather considered as 'data-deficient'

* Scenarios falling under the Null - hypothesis (1 genetic metapopulation or 1 ESU) are **formally tested as a baseline against multi subpopulation- or ESU-scenarios with reversed scoring.**

Figure 1. Lines of evidence (A) and the Two-phase framework (B) to assess the strength of evidence for subpopulations and Evolutionarily Significant Units (ESUs). Different lines of evidence are highlighted in color (dark green, green and blue). Maximum scores are the numbers provided below each category in the boxes based on genetic or non-genetic evidence from the literature and other documentation. Below each phase the scoring system is explained by exemplary imaginary 'scenarios'.

Getting Started

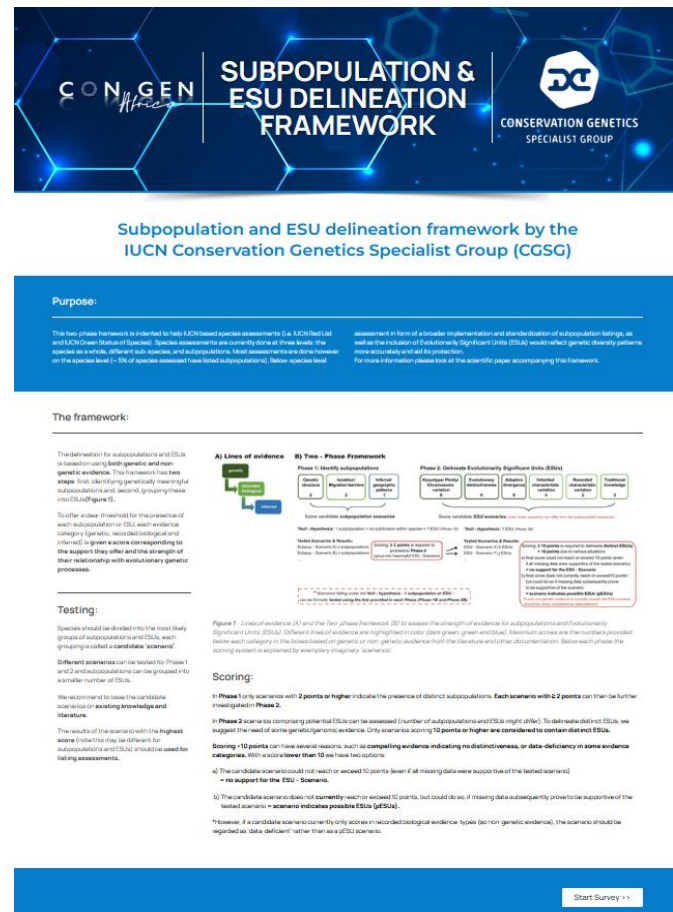
The webtool can be accessed from www.congenafrika.com/survey/.

On the landing page you will get an overview of the framework, including its purpose, different sections of the framework, and its scoring system.

Purpose of the framework

This two-phase framework is intended to help identify appropriate units for biodiversity threat assessments as employed by the IUCN (i.e., IUCN Red List and IUCN Green Status of Species) and other entities. The IUCN currently conducts assessments at three levels: species, subspecies, and subpopulations. The vast majority of assessments are carried out at the species level with subpopulations identified in only 5% of assessments. Expanding these assessments below the species-level through a standardised definition of subpopulations, together with the inclusion of Evolutionarily Significant Units (ESUs), would better capture patterns of genetic diversity and ensure sustainable conservation efforts.

For more information, please look at the scientific paper accompanying this framework (DOI: 10.1093/biosci/biag042).



Overview of the framework

The delineation of subpopulations and ESUs in this framework draws on both genetic and non-genetic evidence. The framework follows a two-tiered approach: first, identifying genetically meaningful subpopulations and, second, testing subpopulations or groups of subpopulations as ESUs (Figure 1).

To provide clear thresholds for recognising each subpopulation or ESU, evidence is categorised as “genetic”, “recorded biological” or “inferred”. A score is assigned to each category of evidence to indicate the relative strength and relevance for determining evolutionary distinctiveness among subpopulations or ESUs.

Scoring system

In **Phase 1** only scenarios with 2 points or higher indicate the presence of distinct subpopulations. Scenarios meeting this threshold can move to Phase 2 to investigate whether the subpopulations or groups of subpopulations qualify as ESUs.

In **Phase 2** scenarios scoring 10 points or higher indicate the presence of ESUs. This can only be achieved with genetic/genomic evidence.

Scores below 10 points may result from either:

- **No support for ESUs.** In this situation, even if missing data were assumed supportive for the tested scenario, a score of 10 could not be reached.
- **Possible ESUs (pESUs).** In this situation, the scenario does not **currently** score 10 points, but could if new evidence later proves to be supportive.

*Note: If the score of a candidate scenario is based **only** on recorded biological (non-genetic) evidence, the scenario should be regarded as having insufficient data, and hence not a pESU.

Species selection

It may not be necessary to assess all species, although there is no restriction either. This framework is intended to help clarify population structure and guide conservation decision-making. Priority should be given to species with likely or suspected genetic structure, to test the relevance of subspecies designations, to understand appropriate management units for translocations, and to taxa that may be experiencing different threats across their range.

Scenario selection

Species should be divided into the most likely groups of subpopulations and ESUs, hereafter referred to as *candidate scenarios*. These scenarios may be derived from the scientific literature or developed using other relevant criteria based on existing knowledge. This framework is intended for users who are familiar with the species' biology, life history traits and other pertinent information. We therefore advise against the exhaustive testing of all possible scenarios (e.g., 1-7+ subpopulations or ESUs), as many are likely to lack sufficient supporting data. Different scenarios can be tested in Phase 1 and Phase 2, as multiple subpopulations may be grouped into a single ESU. As the default, the first scenario is always the null hypothesis (either a single Metapopulation or a single ESU) and is formally tested against other candidate scenarios as a baseline. Here the scoring is reversed to keep the questions identical across all scenarios. To start the assessment, go to www.congenafrika.com/survey/, click the **“Start Survey” button**.

Conducting the survey

Navigating the survey pages

Left-hand information panel

On every testing page, a panel appears on the **left** side of the screen. This panel provides definitions for technical terms, explanations of principles used in the framework, and links to relevant reference materials. It is intended to help clarify concepts and ensure consistent interpretation of the criteria during scoring. Users are encouraged to refer to this panel regularly, especially when encountering unfamiliar terminology or when making decisions that require an understanding of the key principles.

Survey questions and layout

At the start of the survey, users are asked to enter the name of the species and the scenarios being investigated to allow for easier recall and context should an email of the results be requested.

Questions are grouped according to evidence categories ('genetic', 'recorded biological', and 'inferred'). To help users understand the type of information each question requires, information bubbles have been included. Hovering over the 'i' icon, will display additional guidance.

Up to four scenarios can be tested simultaneously, displayed as adjacent columns (1st column is always the null hypothesis, a single metapopulation or ESU). Answers are required for each scenario to enable direct comparison and ensure consistent scoring. If more than 4 scenarios need to be assessed, additional surveys must be completed, and the scores for each scenario should be recorded and retained for reference. This process is supported by the email response function which sends results directly to the user if an email address is provided.

After completing all questions, users have the option to enter additional notes and their email address before submitting the survey. These notes and the email addresses **are not stored or retained** by the survey developers. They are used solely for the user's convenience. Providing your name and email address is optional; however, including this information allows you to receive your

results via email for record-keeping purposes. Clicking the **“Submit Data”** button will display a results screen showing the total scores for each scenario.

Phase 1: Identify subpopulations

To identify meaningful subpopulations, in **Phase 1** the presence of a single metapopulation (1st column) is tested against multi-subpopulation scenarios by using three different evidence categories. Based on existing data and knowledge it is assessed: (1) if **genetic structure** can be determined/detected, (2) if there is **natural isolation of fragmentation** of the subpopulations/metapopulation tested and (3) in case there is no direct (observed) data available of the focal species, if there is **inferred evidence of a likely fragmentation** of the subpopulations/metapopulation.

Phase 1: Identify Subpopulations

This survey is best views on Desktop, or rotate your phone for a better view.

Please use the dropdown menu below for additional information.

- Subpopulations
- Evolutionarily Significant Units (ESU)
- Possible Evolutionarily Significant Unit (pESU)
- Gene Flow
- Potential IUCN Species assessment
- References

Scoring:

- 2 points or higher = distinct subpopulations
- Less than 2 points = stop assessment

Lines of evidence
genetic

Which species are you testing? *

How to set up scenarios to test:
This survey allows the evaluation of **multiple subpopulation scenarios**.

The **first scenario always represents the null hypothesis**, which assumes a **single metapopulation**. This one-metapopulation model must be tested first and serves as the baseline against which all alternative scenarios involving multiple subpopulations are compared.

When selecting scenarios that include **multiple subpopulations** (i.e. not the default metapopulation), **all subpopulations must be mutually distinct and supported by evidence**. Each subpopulation must have a clear evidentiary basis for inclusion in the scenario being tested. If any two subpopulations cannot be distinguished from one another, they should be merged, and a revised scenario should be considered for testing.

How many Scenarios do you want to assess?

4 Scenarios ▼

1 Metapopulation	Scenario B	Scenario C	Scenario D
Scenario 1 name	Scenario 2 name	Scenario 3 name	Scenario 4 name

Remember, only four scenarios can be assessed simultaneously.

Phase 2: Delineate Evolutionarily Significant Units (ESUs)

To delineate subpopulation (groupings), in **Phase 2** the presence of a single ESU (1st column) is tested against multiple ESU scenarios by using six evidence categories. Based on existing data and knowledge it is assessed: (1) if **inherited variation can be detected at the chromosome level** (number, structure or ploidy level), (2) if there is genetic evidence of **long-term reproductive isolation**, (3) if **local adaption** can be detected, if there are **characteristic differences** recorded due to (4) **inheritance** or (5) **transmission** and (6) if there is any indication of distinctives of tested ESUs based on **traditional and local knowledge** (not scientifically tested).

Phase 2: Delineate Evolutionarily Significant Units (ESUs)

Please use the information provided by the email sent, to complete the following ESU survey.

Please use the dropdown menu below for additional information.

- Subpopulations
- Evolutionarily Significant Units (ESU)
- Probable Evolutionarily Significant Unit (pESU)
- Gene Flow
- Potential IUCN Species assessment
- References

Scoring:

- 10 points or higher = distinct ESU
- Below 10 points:
 - a) 10 points cannot be reached or exceeded (even if all missing data were supportive of the tested scenario) = no support for the ESU - Scenario.

Which species are you testing?

How to set up scenarios to test:
This survey allows the evaluation of **multiple ESU scenarios**.

The **first scenario always represents the null hypothesis**, which assumes a **single Evolutionarily Significant Unit (ESU)**. This one-ESU model must be tested first and serves as the baseline against which all alternative scenarios involving multiple ESUs are compared.

When selecting scenarios that include **multiple ESUs** (i.e. not the default single ESU), **all ESUs must be mutually distinct and supported by evidence**. Each ESU must have a clear evidentiary basis for inclusion in the scenario being tested. If any two ESUs cannot be distinguished from one another, they should be merged, and a revised scenario should be considered for testing.

How many scenarios are you testing?

4 Scenarios

1 ESU	Scenario B	Scenario C	Scenario D
Scenario 1 name	Scenario 2 name	Scenario 3 name	Scenario 4 name

Remember, only four scenarios can be assessed simultaneously

Case Study

To assist in navigating the framework, a detailed case study is provided, for which existing literature and previous knowledge was used to determine candidate scenarios which were tested in Phase 1 and 2. We did not intend to test all possible scenarios, since that would result in many “data-deficient” scenarios. We rather built on existing theories and based our scenarios on them.

The Western Leopard Toad (*Sclerophrys pantherina*)

The Western Leopard Toad (WLT) is a narrow endemic toad, localised to a small region of the Western Cape Province of South Africa and genetic as well as morphological studies have been conducted, therefore the WLT represents a good case study.

1. Fill in species and scenario information.

Based on two scientific studies utilising genetic data and the geographic distribution of the WLT, three different candidate scenarios are tested here: (1) a single metapopulation (null hypothesis), (2) two subpopulations representing the Cape Peninsula and the Overstrand region, and (3) three subpopulations, where the northern locality of this species on the Cape Peninsula is further separated from the remaining southern distribution.

Phase 1: Identify Subpopulations

This survey is best views on Desktop, or rotate your phone for a better view.

Please use the dropdown menu below for additional information.

- Subpopulations
- Evolutionarily Significant Units (ESU)
- Possible Evolutionarily Significant Unit (pESU)
- Gene Flow
- Potential IUCN Species assessment
- References

Scoring:

- 2 points or higher = distinct subpopulations
- Less than 2 points = stop assessment

Lines of evidence

Which species are you testing? *

Sclerophrys pantherina

How to set up scenarios to test:

This survey allows the evaluation of **multiple subpopulation scenarios**.

The **first scenario always represents the null hypothesis**, which assumes a **single metapopulation**. This one-metapopulation model must be tested first and serves as the baseline against which all alternative scenarios involving multiple subpopulations are compared.

When selecting scenarios that include **multiple subpopulations** (i.e. not the default metapopulation), **all subpopulations must be mutually distinct and supported by evidence**. Each subpopulation must have a clear evidentiary basis for inclusion in the scenario being tested. If any two subpopulations cannot be distinguished from one another, they should be merged, and a revised scenario should be considered for testing.

How many Scenarios do you want to assess?

3 Scenarios

1 Metapopulation	2 Subpopulations	3 Subpopulations
Scenario 1 name	Scenario 2 name	Scenario 3 name

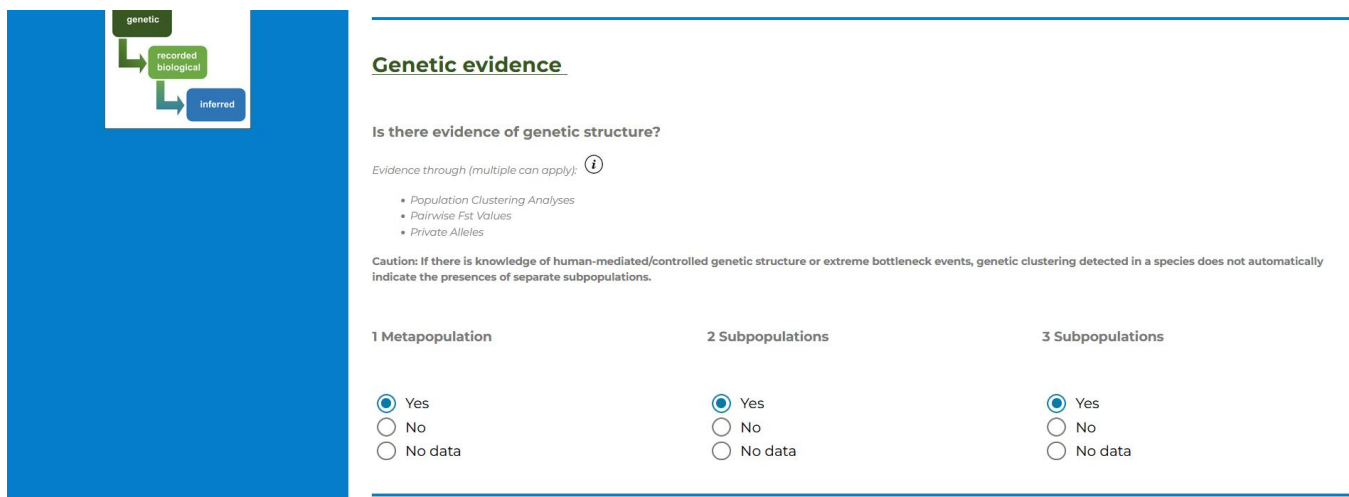
2. Answer the questions for each scenario.

a. Genetic evidence

In **Scenario (1)** (null hypothesis), depicting a single metapopulation, scoring is reserved to keep the questions identical across all scenarios. Mitochondrial DNA and microsatellite data did not support the presence of a single metapopulation, as strong genetic isolation was observed between the Cape Peninsula and the Overstrand region with no shared haplotypes (Measey & Tolley, 2011; J. M. da Silva, unpublished data). Therefore, a score of 0/2 for genetic structure was assigned.

In **Scenario (2)**, mitochondrial ND2 data indicated differentiation between the Cape Peninsula and Overstrand, with no shared haplotypes in the available samples (Casola 2017; Measey and Tolley 2011). This regional separation was also recovered in unpublished microsatellite and ddRAD analyses, which did not support isolation by distance as the primary explanation for the observed pattern (J. M. da Silva, unpublished data). Together, these findings resulted in a score of 2/2 for genetic structure.

In **Scenario (3)**, unpublished microsatellite and ddRAD data indicated a more complex structure with $K = 3$, including differentiation of the northern-most Cape Peninsula locality (Observatory) from both the Overstrand and southern Peninsula localities (J. M. da Silva, unpublished data). Despite this finer-scale subdivision, clear genetic structure was still present, resulting in a score of 2/2 for genetic structure.



The screenshot shows a web form titled "Genetic evidence". On the left is a blue sidebar with a navigation menu containing three items: "genetic" (highlighted in green), "recorded biological", and "inferred". The main content area has the title "Genetic evidence" in green. Below it is the question "Is there evidence of genetic structure?". A note says "Evidence through (multiple can apply):" followed by an information icon and a list of three bullet points: "Population Clustering Analyses", "Pairwise Fst Values", and "Private Alleles". A caution note states: "Caution: If there is knowledge of human-mediated/controlled genetic structure or extreme bottleneck events, genetic clustering detected in a species does not automatically indicate the presences of separate subpopulations." Below this, there are three columns for different population structures: "1 Metapopulation", "2 Subpopulations", and "3 Subpopulations". Each column has three radio button options: "Yes", "No", and "No data". In the "1 Metapopulation" column, "Yes" is selected. In the "2 Subpopulations" column, "Yes" is selected. In the "3 Subpopulations" column, "Yes" is selected.

b. Recorded biological evidence

In **Scenario (1)** (null hypothesis), depicting a single metapopulation, scoring is reserved to keep the questions identical across all scenarios. The existing evidence did not indicate a single metapopulation. Therefore, a score of 0/2 for isolation/migration barriers was assigned.

In **Scenario (2)**, approximate Bayesian inference and spatial analysis of molecular variance indicate historical bottleneck events associated with sea level changes during interglacial periods, which likely isolated the Cape Peninsula and Overstrand subpopulations (Casola 2017; Measey and Tolley 2011). In addition, the historical loss of breeding sites further supports long-term isolation (Measey and Tolley 2011). Therefore, a score of 2/2 for isolation/migration barriers was assigned.

In **Scenario (3)**, although genetic structure exists between the Observatory locality and southern Peninsula on the Cape Peninsula, this is interpreted as human-mediated due to the presence of major highways which began construction in the 1950s, which restrict gene flow for approximately 30+ generations. Therefore, this scenario does not reflect natural isolation processes and received a score of 0/2 for isolation/migration barriers.

Recorded biological evidence

Is there evidence of natural isolation or subpopulation fragmentation (through migration barriers)?

Evidence through (multiple can apply):

- Occupation of different biogeographical zones
- Occupation of discrete remnants of historical habitats with little/no natural migration

Caution: In this framework, the natural fragmentation of subpopulations should be distinguished from human-mediated/controlled fragmentation of subpopulations, since the latter does not automatically indicate the presences of separate subpopulations.

1 Metapopulation	2 Subpopulations	3 Subpopulations
☒ Yes ☐ No ☐ No data	☒ Yes ☐ No ☐ No data	☐ Yes ☒ No ☐ No data

c. Inferred evidence

In **Scenario (1)** (null hypothesis), depicting a single metapopulation, scoring is reversed to keep the questions identical across all scenarios. Species distribution models across the range of WLT indicate a clear climatic and habitat suitability disjunction between the Cape Peninsula and Overstrand subpopulations (Casola, 2017). This existing evidence did not indicate a single metapopulation. Therefore, a score of 0/2 for inferred evidence was assigned.

In **Scenario (2)**, the same modelled disjunction between the Cape Peninsula and Overstrand subpopulations supports inferred geographic separation (Casola, 2017) and indicates two distinct subpopulations. This resulted in a score of 2/2 for the inferred evidence.

In **Scenario (3)**, no gaps in climatic suitability were detected within the Cape Peninsula between the southern Peninsula and the northern Observatory site, indicating no inferred barrier at this finer scale. Therefore, this scenario received a score of 0/2 for inferred evidence was assigned.

Inferred evidence

Is there inferred evidence of likely subpopulation fragmentation?

Note: Only relevant if genetic or recorded evidence from the focal species is absent. Evidence of inferred patterns can be used, if nothing is known about the focal species, co-occurring or closely related species could give indication of subpopulation structure.

If genetic or recorded evidence (see Questions above) for the focal species is available, select 'Not applicable' and continue to the results/next phase.

Evidence through (multiple can apply): ⓘ

- **inferred** subpopulation disjunction (e.g., species distribution modeling)
- Phylogeographic and/or biogeographic evidence **from co-occurring species**
- Assumption of biogeographical data/refugia **without direct evidence**
- **inferred** dispersal distance and buffer according to the CBD indicators

1 Metapopulation	2 Subpopulations	3 Subpopulations
☒ Yes	☒ Yes	☐ Yes
☐ No	☐ No	☒ No
☐ No data	☐ No data	☐ No data
☐ Not applicable	☐ Not applicable	☐ Not applicable

3. Provide your name and e-mail address (optional). This will allow the results to be sent to your email address.

Notes

Name & Surname
Jane Doe

Email Address
jane.doe@xxxxx

Data protection disclaimer
By providing your name and email, you will receive the results in your inbox, as well as on the next page and may receive communications from us to participate further in the development of this framework. If you **choose not to enter your email**, your results will still be presented to you on the next page.

Submit Data

4. Click the **“Submit Data”** button to receive the results (mandatory to get results).

5. Interpreting the results.

This is the results page. It lists each question and the responses entered by the user as verification. Below the questions is a summary of the scores for each scenario.

For the Western Leopard Toad, two of the three candidate scenarios (**Scenarios (2) and (3)**) meet the minimum threshold of two points, and therefore qualify as subpopulation scenarios. The final tallies for Phase 1 however indicate that **Scenario (2)** (two subpopulations) received the highest score, suggesting that the presence of two naturally fragmented subpopulations is best supported. From a management perspective, **Scenario (3)** should still be considered, as the Observatory population harbours unique genetic diversity that warrants protection. Given that re-establishing natural movement corridors between Observatory and the southern Peninsula is unlikely, management efforts should focus on maintaining this isolated population.

Phase 1: Identify Subpopulations

This survey is best views on Desktop, or rotate your phone for a better view.

Please use the dropdown menu below for additional information.

- ▶ Subpopulations
- ▶ Evolutionarily Significant Units (ESU)
- ▶ Possible Evolutionarily Significant Unit (pESU)
- ▶ Gene Flow
- ▶ Potential IUCN Species assessment
- ▶ References

Scoring:

- 2 points or higher = distinct subpopulations
- Less than 2 points = stop assessment

Lines of evidence

genetic

↓

recorded biological

Thank you , for participating in the **CGSG Subpopulation and ESU survey**.

The species you tested is the **Sclerophrys pantherina**

Your Survey Results:

1. Is there evidence of population structure? (2 points)

1 Metapopulation Yes	2 Subpopulations Yes	3 Subpopulations Yes	
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2. Is there evidence of natural isolation or subpopulation fragmentation (through migration barriers)? (2 points)

1 Metapopulation Yes	2 Subpopulations Yes	3 Subpopulations No	
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3. Is there inferred evidence of likely subpopulation fragmentation? (1 points)

1 Metapopulation Yes	2 Subpopulations Yes	3 Subpopulations No	
-------------------------	-------------------------	------------------------	--

Total score for the scenario: **1 Metapopulation** is 0
Total score for the scenario: **2 Subpopulations** is 5
Total score for the scenario: **3 Subpopulations** is 2
Total score for the scenario: is 0

For scenarios that scored 2 points or higher, please use the button below to continue to Phase 2.

GO TO PHASE 2 >>

You'll be receiving the results in your inbox soon!

Both scenarios 2 and 3 qualify for testing in Phase 2.

6. Click on the “Phase 2” button to proceed.

7. As in Phase 1, enter the species and scenario information. This does not get carried over from Phase 1.

Based on the results of Phase 1, we tested three candidate scenarios for delineating Evolutionarily Significant Units (ESUs): (1) a single ESU, because subpopulations can be nested within an ESU (2) two ESUs (Cape Peninsula and Overstrand), (3) three ESUs (Observatory, southern Peninsula of Cape Town and Overstrand).

Phase 2: Delineate Evolutionarily Significant Units (ESUs)

Please use the information provided by the email sent, to complete the following ESU survey.

Please use the dropdown menu below for additional information.

- ▶ Subpopulations
- ▶ Evolutionarily Significant Units (ESU)
- ▶ Probable Evolutionarily Significant Unit (pESU)
- ▶ Gene Flow
- ▶ Potential IUCN Species assessment
- ▶ References

Scoring:

- 10 points or higher = distinct ESU
- Below 10 points:
 - a) 10 points cannot be reached or exceeded (even if all missing data were supportive of the tested scenario) = no support for the ESU – Scenario.

Which species are you testing?

Sclerophrys pantherina

How to set up scenarios to test:
This survey allows the evaluation of **multiple ESU scenarios**.

The **first scenario always represents the null hypothesis**, which assumes a **single Evolutionarily Significant Unit (ESU)**. This one-ESU model must be tested first and serves as the baseline against which all alternative scenarios involving multiple ESUs are compared.

When selecting scenarios that include **multiple ESUs** (i.e. not the default single ESU), **all ESUs must be mutually distinct and supported by evidence**. Each ESU must have a clear evidentiary basis for inclusion in the scenario being tested. If any two ESUs cannot be distinguished from one another, they should be merged, and a revised scenario should be considered for testing.

How many scenarios are you testing?

3 Scenarios ▼

1 ESU	2 ESUs	3 ESUs
<i>Scenario 1 name</i>	<i>Scenario 2 name</i>	<i>Scenario 3 name</i>

8. Answer the questions for each line of evidence for each scenario being investigated.
 - a. Genetic evidence

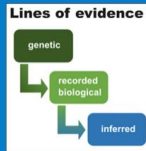
In **Scenario (1)** (null hypothesis), depicting a single ESU, scoring was reserved to maintain consistency across scenarios. No chromosomal or karyotypic information is available for this species, resulting in 0/6 for karyotype variation. Mitochondrial DNA and population-level analyses indicate strong genetic isolation between the Cape Peninsula and Overstrand regions, with no shared haplotypes (Measey and Tolley 2011). ABC and SAMOVA analyses have indicated that several bottleneck events associated with sea level rise during interglacial periods, contributed to the isolation of the Cape Peninsula and Overstrand distributions (Casola 2017; Measey and Tolley 2011), contradicting a single ESU and resulting in 0/6 for evolutionary distinctiveness. No evidence for adaptive divergence data is available for this scenario (0/6 points).

In **Scenario (2)**, representing two ESUs, chromosomal data remain unavailable (0/6 points). Genetic evidence still shows strong isolation between the Cape Peninsula and Overstrand regions, supported by mitochondrial DNA and bottleneck analyses (Casola 2017; Measey and Tolley 2011), resulting in 6/6 for evolutionary distinctiveness. No evidence for adaptive divergence data is available for this scenario (0/6 points).

In **Scenario (3)**, no chromosomal data were available (0/6 points). Mitochondrial analyses placed the Observatory within the broader Cape Peninsula group (Casola 2017; Measey and Tolley 2011). Therefore 0/6 points was assigned for evolutionary distinctiveness. No adaptive divergence data are available for this scenario (0/6 points).

0/10 points are currently not reached or exceeded, but could if missing data subsequently prove to be supportive of the tested scenario = scenario indicates possible ESUs (pESUs)*

*However, if a candidate scenario currently only scores in recorded biological evidence-types (so non-genetic), the scenario should be regarded as 'data-deficient' rather than as a pESU scenario.



Genetic evidence

Is there inherited variation in chromosome numbers, ploidy level or chromosome structure?

Evidence through (multiple can apply):

- Karyotype variation
- Difference in ploidy levels between units
- Structural variation of chromosomes between units

1 ESU

- ☐ Yes
☐ No
☒ No data

2 ESUs

- ☐ Yes
☐ No
☒ No data

3 ESUs

- ☐ Yes
☐ No
☒ No data

Is there evidence of long-term reproductive isolation?

Evidence through (multiple can apply):

- Reciprocal Monophyly
- Molecular Estimate of Divergence Time

1 ESU

- ☒ Yes
☐ No
☐ No data

2 ESUs

- ☒ Yes
☐ No
☐ No data

3 ESUs

- ☐ Yes
☒ No
☐ No data

Is there evidence of local adaptation or adaptive divergence?

Evidence through (multiple can apply):

- Genomic signals of local adaptation
- Evidence from hybrid zones

1 ESU

- ☐ Yes
☐ No
☒ No data

2 ESUs

- ☐ Yes
☐ No
☒ No data

3 ESUs

- ☐ Yes
☐ No
☒ No data

b. Recorded biological evidence

In **Scenario (1)** (null hypothesis), depicting a single ESU, scoring was reserved to maintain consistency across scenarios. WLTs exhibit strong breeding site fidelity, with adults returning annually to the same breeding wetlands and ponds (late July to September). This inherited behavioural trait may reinforce population structuring and limit dispersal and is contradicting a single ESU and resulting in 0/4 for inherited characteristic variation. No evidence was found for recorded variation or traditional knowledge in either scenario (0/2 each).

In **Scenario (2)**, strong breeding site fidelity is observed across the species, with adults returning to the same water bodies each breeding season, potentially reinforcing population structure. However, no evidence links this behaviour to distinct ESU-level differences. This information was nevertheless considered because breeding site fidelity is an inherited behavioural trait with the potential to influence genetic differentiation and local adaptation. Since the behaviour appears consistent across the species rather than

varying among ESUs, it did not provide support for distinguishing separate units and was therefore scored as 0/4 for inherited characteristic variation. No evidence was found for recorded variation or traditional knowledge in either scenario (0/2 each).

In **Scenario (3)**, the same strong breeding site fidelity is present across populations, but there is no evidence that this behaviour differs among finer-scale genetic subdivisions. This information was nevertheless considered because site fidelity can contribute to restricted dispersal and fine-scale structuring, even when not directly linked to adaptive differences. However, as the behaviour was shared broadly and did not distinguish among the proposed subpopulations, it did not support subdivision at this level and was therefore scored as 0/4 for inherited characteristic variation. No evidence was found for recorded variation or traditional knowledge in either scenario (0/2 each).

Recorded biological evidence

Is there evidence of inherited characteristic differences?

Evidence through (multiple can apply): 

- Heritable differences in focal traits observed in experiments or confirmed by analytical tests

1 ESU

- ☒ Yes
☐ No
☐ No data

2 ESUs

- ☒ Yes
☐ No
☐ No data

3 ESUs

- ☒ Yes
☐ No
☐ No data

Is there evidence supporting transmitted characteristic differences?

Evidence through (multiple can apply): 

- Cultural or learnt behavioral differences unique or specific between units
- Acquired or transmitted traits (e.g., migration pattern, methylation differences, body size differences)

1 ESU

- ☐ Yes
☐ No
☒ No data

2 ESUs

- ☐ Yes
☐ No
☒ No data

3 ESUs

- ☐ Yes
☐ No
☒ No data

Is there traditional knowledge suggesting distinctiveness between units?

Evidence through (multiple can apply): 

- Recorded differences based on traditional and local knowledge

1 ESU

- ☐ Yes
☐ No
☒ No data

2 ESUs

- ☐ Yes
☐ No
☒ No data

3 ESUs

- ☐ Yes
☐ No
☒ No data

9. Enter your name and e-mail address (optional).



Notes

Name & Surname

Jane Doe

Email Address

jane.doe@xxxxx

Data protection disclaimer

By providing your name and email, you will receive the results in your inbox, as well as on the next page and may receive communications from us to participate further in the development of this framework. If you **choose not to enter your email**, your results will still be presented to you on the next page.

Submit Data

10. Click the “**Submit Data**” button to receive the results.

11. Interpreting the results.

The results of Phase 2 clearly identify two ESUs within the Western Leopard Toad (**Scenario (2)**; 10 points). The other scenarios were not supported, with **Scenario (1)** (a single ESU) receiving a score of zero and **Scenario (3)** a score of four points. As these scores are below the threshold of 10 and include ‘No data’ responses, **Scenario (3)** remains plausible (i.e., 3 pESUs). Should additional data become available, **Scenario (3)** should be re-examined.

Phase 2: Delineate Evolutionarily Significant Units (ESUs)

Please use the information provided by the email sent, to complete the following ESU survey.

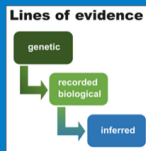
Please use the dropdown menu below for additional information.

- ▶ Subpopulations
- ▶ Evolutionarily Significant Units (ESU)
- ▶ Probable Evolutionarily Significant Unit (pESU)
- ▶ Gene Flow
- ▶ Potential IUCN Species assessment
- ▶ References

Scoring:

- 10 points or higher = distinct ESU
- Below 10 points:
 - a) 10 points cannot be reached or exceeded (even if all missing data were supportive of the tested scenario) = **no support for the ESU - Scenario.**
 - b) 10 points are **currently** not reached or exceeded, but could if missing data subsequently prove to be supportive of the tested scenario = **scenario indicates possible ESUs (pESUs)***

*However, if a candidate scenario currently only scores in recorded biological evidence-types (so non-genetic), the scenario should be regarded as 'data-deficient' rather than as a pESU scenario.



Thank you , for participating in the **CGSG Subpopulation and ESU survey**.

The species you tested is the **Sclerophrys pantherina**

Your Survey Results for Phase 2:

1. Is there inherited variation in chromosome numbers, ploidy level or chromosome structure? (6 points)

1 ESU No data	2 ESUs No data	3 ESUs No data	
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2. Is there evidence of long-term reproductive isolation? (6 points)

1 ESU Yes	2 ESUs Yes	3 ESUs No	
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3. Is there evidence of local adaptation or adaptive divergence? (6 points)

1 ESU No data	2 ESUs No data	3 ESUs No data	
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4. Is there evidence of inherited characteristic differences? (4 points)

1 ESU Yes	2 ESUs Yes	3 ESUs Yes	
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5. Is there evidence supporting transmitted characteristic differences? (2 points)

1 ESU No data	2 ESUs No data	3 ESUs No data	
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6. Is there traditional knowledge suggesting distinctiveness between units? (2 points)

1 ESU No data	2 ESUs No data	3 ESUs No data	
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Total score for the scenario: **1 ESU** is 0
 Total score for the scenario: **2 ESUs** is 10
 Total score for the scenario: **3 ESUs** is 4
 Total score for the scenario: is 0

Scenario scoring:

- Scenarios scoring 10 points or higher = distinct ESUs
- Scenarios scoring below 10 points:
 - **without** potential to reach/exceed 10 points = no support for tested scenario
 - **with** potential to reach/exceed 10 points = possible ESU (pESU) status

You'll be receiving the results in your inbox soon!

You've reached the end of the survey. We trust that the information will provide insight into the management of your species.

Reference for Case Study

Casola, Sarah. 2017. *The Potential Impact of Climate Change on the Genetic Diversity of the Endangered Western Leopard Toad, Sclerophrys Pantherina*.

<http://hdl.handle.net/11427/25012>.

Measey, G. John, and Krystal A. Tolley. 2011. "Investigating the Cause of the Disjunct Distribution of Amietophrynus Pantherinus, the Endangered South African Western Leopard Toad." *Conservation Genetics* 12 (1): 61–70. <https://doi.org/10.1007/s10592-009-9989-7>.

FAQs

Understanding the framework

What's the difference between a subpopulation and an ESU?

Subpopulations as used here most closely relate to the IUCN's classification, whereby population refers to the species as a whole, and subpopulations represent demographic groupings (often with limited gene flow), which others may refer to as populations.

ESUs (or Evolutionarily Significant Units) on the other hand are evolutionary lineages that typically reflect deeper genetic and ecological divergence. Should we state here which definition we used? There are multiple definitions so would be good to be clear here.

Why is there the option to test a single metapopulation or ESU?

While users can infer the rejection of all multi-group scenarios to mean the lack of structure and hence a single metapopulation or ESU for the species as a whole, others may wish to systematically assess this. This can be particularly useful when testing the validity of an existing designation (e.g., a recognised subspecies or management unit).

Since the logic of the questions differs when testing for a single grouping (often reversing the framing of comparison questions), a separate survey path is provided to ensure clarity and accuracy in scoring.

How does this framework align with the current IUCN process and international agreements such as the Kunming-Montreal Global Biodiversity Framework?

This framework complements and strengthens existing IUCN processes by providing a more standardised and biologically grounded method for identifying intraspecific units—such as subpopulations and Evolutionarily Significant Units (ESUs)—that may warrant separate consideration in Red List or Green Status assessments. While the IUCN permits assessments below the species level (e.g., subspecies and subpopulations), the criteria for defining these units are often inconsistently applied. This framework addresses that gap by offering clear, evidence-based guidance to support more consistent and defensible decisions.

In the broader global context, the framework is also closely aligned with international policy targets, such as Target 4 of the Kunming-Montreal Global Biodiversity Framework (GBF), which emphasises the importance of preventing extinction, halting population declines, and maintaining genetic diversity within species. It supports this target by helping users recognise and conserve genetically meaningful units (subpopulations correlates to populations in GBF context), which is essential for long-term species resilience and adaptive potential.

By improving how we recognise and assess intraspecific diversity, the framework contributes directly to the implementation of the IUCN Red List Guidelines and the goals of the GBF, particularly around conservation planning, recovery prioritization, and tracking progress on the state of biodiversity at national and global levels.

Using the framework and webtool

How do I know which scenarios to test?

Start by identifying plausible ways that the species might be divided in subpopulations or ESUs, based on any existing information. Scenarios can be drawn from:

- Published scientific literature
- Expert knowledge or field observations
- Known geographic or ecological barriers
- Morphological or behavioural differences
- Suspected areas of genetic divergence
- Need to verify/validate subspecific designations

You do not need perfect evidence to define a scenario. This framework is designed to help *test* these possibilities. Try to focus on the most likely or relevant groupings, especially those that might face different threats, show signs of isolation, or are important for management decisions (e.g., translocation units).

You can test up to **four scenarios at a time**, and more can be assessed by completing additional surveys. Each scenario represents a hypothesis about how population structure may be organised.

What should I do with the results?

This framework was developed to help researchers, practitioners and government officials identify and assess genetically meaningful units to inform conservation management and planning. While its application may be broad, we feel it is especially relevant to IUCN assessments (i.e., Red List, Green Status). These assessments are primarily conducted at the species level; however, subspecies and subpopulation assessments are also permitted, but their use is limited and inconsistent.

Subspecies designations are often questioned due to modern genetic evidence revealing little or no support for traditional classifications. Similarly, only around 5% of IUCN assessments include information on subpopulation number or size, and there is considerable variation in how subpopulations are defined and delimited.

This framework was therefore created to help standardise the classification of intraspecific units in a way that is more biologically and evolutionarily meaningful. The results of the survey can guide the appropriate level at which assessments should be conducted—particularly when different subpopulations or Evolutionarily Significant Units (ESUs) face different threats or extinction risks. More broadly, however, the results can strengthen species conservation planning by providing a clearer, evidence-based understanding of population structure.

For scenarios showing data deficiency (i.e., subpopulation scenarios scoring less than 2 or ESU scenario below 10), where population structure is suspected but not well supported, the results can serve as a basis for considering follow-up research (e.g., collecting genetic data).

In all cases, it is highly recommended that users retain all scenario scores and notes. This promotes transparency, facilitates the tracking of decision-making, and allows for reassessments as new data become available.

What if I'm unsure how to score a question?

If you're unsure how to answer a question, use the information bubbles (hover over the “i” icon) for guidance on what kind of evidence is relevant. If uncertainty remains, it's better to select “uncertain” or leave the score blank rather than guessing, as inaccurate scoring can affect the framework's overall result. You may also want to make a note explaining the uncertainty to support transparency and future reassessment.

What if I have evidence based on different genetic markers?

Molecular data has a multitude of forms, ranging from mitochondrial or candidate gene haplotypes, to microsatellite markers, and Single Nucleotide Polymorphisms (SNPs, which themselves can range from hundreds to millions in count). Each marker type provides information on the presence of a subpopulation or an ESU in a species but with different resolution. Higher resolution genomic tools are always likely to find more structure because there are more data available. This however does not always translate into biologically meaningful subpopulation divisions within species. Study design is also key, and can lead to various outcomes/conclusion, when e.g., comparing genomic markers with mitochondrial haplotypes or microsatellite markers, or comparing a range-wide study to a study only using very few sites.

If different genetic markers point to different evidence you may want to assess these evidences in different candidate scenarios.

Can I use this framework for species without genetic data?

Yes. The framework is designed to accommodate a wide range of evidence types, including ecological, morphological, behavioural, and geographic data. While genetic data generally carry the highest weight in determining population structure due to the ability to directly reveal patterns

of gene flow, divergence, and reproductive isolation, the tool can still be meaningfully applied where such data are lacking.

In such cases, users are encouraged to use other relevant indicators of structure, such as:

- Known barriers to dispersal (e.g., rivers, mountains)
- Habitat discontinuities or fragmentation
- Local adaptation or environmental differences
- Morphological variation or behavioural distinctions
- Expert opinion or unpublished field observations

The scoring framework reflects the relative strength of different data types, with genetic evidence typically contributing most strongly to higher scores. However, moderate or even low scores derived from non-genetic data can still be valuable for identifying candidate units and prioritising further investigation. The framework is especially useful for highlighting areas of uncertainty and guiding decisions on where additional data collection (including genetics) may be warranted.

Are there risks of oversplitting or undersplitting species in this framework?

Yes. As with any method used to assess population structure, there is a potential risk of either oversplitting (dividing a species into more units than is biologically justified) or undersplitting (failing to recognise meaningful structure). This framework aims to reduce those risks by offering a transparent, repeatable, and evidence-based approach.

The structured questions and weighted scoring system help users assess the strength of evidence for population or evolutionary distinctiveness. Genetic evidence carries the highest weight because it provides direct insights into gene flow and divergence, but other data types are also considered. By integrating multiple lines of evidence, the framework promotes balanced decision-making.

To further minimise misclassification:

- Users are encouraged to test only plausible scenarios grounded in available data.
- The scoring system accommodates uncertainty (e.g., “uncertain” responses don’t contribute to the total).
- The framework does not prescribe taxonomic action—it is a decision-support tool and does not replace formal taxonomic classification or phylogenetic studies.

In short, while some subjectivity is unavoidable, the framework helps make assumptions and limitations explicit, reducing arbitrary or inconsistent decisions and supporting more defensible classifications.

Survey functionality

Can I go back and edit a scenario after submitting the survey?

Unfortunately, no. Survey responses are not stored on a database and therefore they cannot be retrieved or edited once submitted. However, if you provide your email address before submission, a copy of the questions, answers and scoring results will be sent to you. This allows you to review the tested scenarios later, consider exploring additional scenarios, or update the assessment if new information becomes available.

Can I save my progress and return later?

Unfortunately, no. Surveys cannot be saved and must be completed in a single session.

What format will the results be in?

The results are sent as plain text in an email, summarising the scores for each tested scenario along with any notes you provided. Please note that results are not sent as PDF or Word (.doc) files.

Can I compare results across surveys?

Yes. Because each scenario receives a discrete score, results for the same species can be meaningfully compared across surveys. Each survey allows testing up to four scenarios, so completing additional surveys enables broader comparisons. While the tool does not automatically aggregate or analyse results, users can manually review and compare saved email summaries.

Can multiple users collaborate on a survey?

Yes, but only if working together in person. The survey can only be completed and submitted from a single computer, as progress cannot be saved or shared across devices. Collaborative input is encouraged, but responses must be entered and submitted for one session.

Are my results or email address stored?

No. All content entered in the survey is not stored or retained by the developers. The email address option provided at the end of each survey is simply available should the user want a saved copy of the survey questions, answers and results (scores) to refer to in future.

Contact Support

Should you have any questions regarding the functionality, use and interpretation of the survey, please do not hesitate to contact ConGen Africa at info@congenafrica.com.